

JOY TIMMERMANS

AI & Computer Vision Engineer

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 4+ years of experience delivering computer-vision, data-platform, and edge-AI systems from prototype through production. Built systems that reduced false positives by 30%, cut cloud costs by 15%, improved real-time inference speed by 2.5×, and removed manual device-provisioning work at scale. Strong in data-centric AI, embedded development, MLOps, and turning technically difficult requirements into working production systems.

EXPERIENCE

LeftFields

AI Engineer

Hasselt, Belgium

February 2026 – Present

- **Rematics:** Owned work across dataset workflows, training infrastructure, and secure edge-AI deployment.
 - * Built source-to-data-lake workflows that reduced dataset preparation from days or even weeks to hours.
 - * Used a hardware security chip to create a fully unsupervised provisioning system, eliminating 30–40 minutes of manual setup per device easily saving 8000 hours a year.
 - * Improved deployment reliability across Yocto-based operating systems and NVIDIA DeepStream pipelines.
- **Daymaker:** Built the product-usage reporting layer used to make traffic and adoption data actionable.
 - * Designed and built an analytics dashboard covering user traffic, product usage, and key behavioural flows.
 - * Turned scattered product data into reporting views that supported day-to-day product and operational decisions.
- **RapidFit:** Audited and redesigned internal workflows across quotation and operation systems.
 - * Mapped existing business and engineering processes to identify bottlenecks, manual work, and tooling gaps.
 - * Proposed process changes that reduced selected business-process turnaround times by up to 60%.

Padelize

CTO

Hasselt, Belgium

July 2026 – Present

- **Technical ownership:** Own the architecture and technical direction for the mobile application, cloud infrastructure, AI services, and deployment pipeline.
- **Computer vision systems:** Lead development of temporal models, field detection, tracking, and homography-based player projection for padel analysis.
- **Model performance:** Reworked the field-detection pipeline from unusable output to 90% accuracy while reaching 2.4× real-time inference speed.
- **Data quality:** Set annotation standards, review dataset quality, and run feedback loops from model failures back into labelling and training.
- **Delivery:** Set technical priorities, convert product requirements into implementation milestones, and unblock engineering delivery across the team.

Secury360

Machine Learning Engineer

Zonhoven, Belgium

August 2022 – February 2026

- **Computer-vision data platform:** Designed and maintained FiftyOne-based pipelines for multi-million-sample datasets, supporting dataset inspection, curation, and model iteration.
- **False-positive reduction:** Applied data-centric AI methods to improve dataset quality and reduce false positives by 30%.
- **Temporal vision modelling:** Developed a preprocessing method that encoded temporal information into stationary image representations for downstream vision models. Increasing the model accuracy while reducing the false positive rate by 40%.
- **Cloud-cost reduction:** Built Azure-based AI filtering logic that reduced monthly cloud costs by 15%.
- **Production edge systems:** Owned Linux-based edge-device software for connectivity, health monitoring, displaying information and reliable deployment in production environments.

PXL Research

Intern Researcher — Time Series & Edge AI

Hasselt, Belgium

February 2022 – July 2022

- **Digital Energy:** Contributed to industrial energy-optimisation research focused on predictive maintenance and real-time monitoring.
- **Forecasting models:** Built multivariate time-series models that forecast 24-hour power consumption with 85% accuracy.
- **Edge constraints:** Evaluated edge-deployment limitations for real-time, sensor-driven industrial energy management.

PXL Smart-ICT

React Native Developer

Hasselt, Belgium

July 2020 – May 2021

- **Healthcare communication platform:** Built a React Native application that streamlined communication between healthcare institutions and workers.

NYBE

Junior Developer

Hasselt, Belgium

May 2022 – July 2022

- **Web development:** Contributed to scalable Laravel applications and production web-development workflows.

EDUCATION

Woolf University

Master of Science in Artificial Intelligence

Remote

2026 – Expected 2027

- **Focus areas:** Computer Vision, Machine Learning, Agentic AI, Statistics for Data Analysis.

PXL University of Applied Sciences and Arts

Bachelor of Applied Computer Science — AI & Robotics

Hasselt, Belgium

2019 – 2022

- **Honors:** Graduated with Honors
- **Relevant coursework:** Machine Learning, Robotics and Path Planning, Big Data, Large Language Models, and Time Series Analysis.

SELECTED PROJECTS

Self-Driving Bot — Robotics & Optimisation: Built a ROS-based autonomous robot using neural networks for lateral control, sensor processing, real-time inference, and path optimisation. Earned the first perfect score, 20/20, in the course history. [GitHub repository](#).

Chess Assistant — Computer Vision: Built a computer-vision system that recognises chessboard states and recommends moves through a self-developed chess engine. [GitHub repository](#).

OPEN SOURCE & COMMUNITY

FiftyOne Community: Helped setup the Voxel51 community Discord server and built a custom bot that automated community interactions. [GitHub repository](#).

FiftyOne Contributions: Contributed PyTorch device flexibility and CVAT organisation support to FiftyOne, improving device compatibility and enterprise annotation workflows. [PR #2843](#); [PR #2716](#).

Speaking: Selected as one of four speakers at the [Brussels AI Meetup](#).

TECHNICAL SKILLS

Languages: Python, JavaScript, C, C++, C#, Java, Kotlin

AI & CV: PyTorch, TensorFlow, Tinygrad, FiftyOne, Weights & Biases, Pandas, CVAT, Ultralytics, Roboflow

LLMs & Agentic AI: RAG, LangChain, LangGraph, MCP, LlamaIndex, Ollama, Hugging Face, Qdrant, Whisper

Edge & Inference: ONNX, TensorRT, DeepStream, ROS 2, Yocto, Embedded Linux

Video & Streaming: OpenCV, FFmpeg, GStreamer

Engineering & Cloud: AWS, Azure, Docker, Git, Bash, GitHub Actions, FastAPI, Django, React Native, Laravel, SQLite, MongoDB

Other: Unity, Gazebo, L^AT_EX, Figma

LANGUAGES

Dutch: Native

English: Fluent

French: Elementary